Software Project Management (SPM) (CSC415)

Course Instructor
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Course Overview and Objective

About Software Project Management

- This course familiarizes with different concepts of software project management mainly focusing on:
  - Project Analysis
  - Scheduling
  - Resource Allocation
  - Risk Analysis Monitoring and Control
  - Software Configuration Management.

Objectives of SPM

- The main objective of the course is to provide:
  - Knowledge of different concepts of software project management so that one will be able to understand and handle various projects.
  - Skill to deal with high risk and innovative projects using different project management skills.

Syllabus

Course Contents:

Unit 1: Introduction to Software Project Management (5 Hrs.)

Software engineering problem and software product, software product attributes, Definition of a Software Project (SP), SP Vs. other types of projects activities covered by SPM, categorizing SPs, Project management cycle, SPM framework, types of project plan.

Unit 2: Project Analysis (8 Hrs.)

Introduction, strategic assessment, technical assessment, economic analysis: Present worth, future worth, annual worth, internal rate of return (IRR) method, benefit-cost ratio analysis, including uniform gradient cash flow and comparison of mutually exclusive alternatives.

Unit 3: Activity Planning and Scheduling (7 Hrs.)

Objectives of activity planning, Work breakdown structure, Bar chart, Network planning model: Critical path method (CPM), Program evaluation and review technique (PERT), Precedence diagramming method (PDM), Shortening project duration, Identifying critical activities.

Syllabus

Unit 4: Risk Management (4 Hrs.)
Introduction, nature and identification of risk, risk analysis, evaluation of risk to the schedule using Z-values.

Unit 5: Resource Allocation (4 Hrs.)
Identifying resource requirements, resource allocation, resource smoothening and resource balancing.

Unit 6: Monitoring and Control (4 Hrs.)
Introduction, collecting data, visualizing progress, cost monitoring, earned value analysis, project control.

Syllabus

Unit 7: Managing Contracts and people (5 Hrs.)

Introduction, types of contract, stages in contract, placement, typical terms of a contract, contract management, acceptance, Managing people and organizing terms: Introduction, understanding behavior, organizational behavior: a back ground, selecting the right person for the job, instruction in the best methods, motivation, working in groups, becoming a team, decision making, leadership, organizational structures, conclusion, further exercises.

Unit 8: Software quality assurance and testing (5 Hrs.)

Testing principles and objectives, test plan, types and levels of testing, test strategies, program verification and validation, software quality, SEI-CMM,SQA activities, QA organization structure, SQA plan.

Unit 9: Software Configuration Management (3 Hrs.)

Introduction, need, basic configuration, management function, baseline, configuration management responsibilities.

Syllabus

Laboratory / Project Work:

Students should prepare a project report using different concepts of software project management. The project can be done in groups with at most four members in each group. Each group can select a case study and apply the concepts of software project management focusing on project analysis, scheduling, risk analysis, resource allocation, testing.

Text Book:

1. Software Project Management by Bob Hughes and Mike Cotterell, Latest Publication

Reference Books:

1. "Introduction to Software Project Management & Quality Assurance", Darrel Ince, I. Sharp, M. Woodman, Tata McGraw Hill
2. "Software Project Management: A Unified Framework", Walker Royce, Addison-Wesley, An Imprint of Pearson Education
3. "Managing the Software Process", Watts S. Humphrey, Addison-Wesley, An Imprint of Pearson Education

Evaluation Process

- Attendance
- Assignments
- Term Examinations
- MCQ
- Presentations or Report Submissions
- Project Works

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Chapter 1: Introduction to Software Project Management

Introduction

- Software Engineering is associated with development of software product using well-defined scientific principles, methods and procedures.
- The outcome of software engineering is an efficient and reliable software product.
- An initial definition of Software Engineering is :
    - *"the establishment and use of sound engineering principles in order to obtain, economical software that is reliable and works efficiently on real machines"*
- Most problems are large and sometimes tricky to handle, especially of they present something new that has never been solved before.

Introduction

- So, we must begin investigating it by analyzing it, i.e. by breaking problem into pieces that we can understand and deal with.
- Once we have analyzed the problem, synthesis is done to put pieces together to get solution of the entire problem.
- Thus, problem-solving technique must have two parts:
  - *Analyzing*
  - *Synthesizing*

Introduction

- Software Engineers use tools, techniques, procedures and paradigms (Patterns) to enhance the quality of their software products.
- Their aim is to use efficient and productive approaches to generate effective solutions to problems.
- Basically Software Engineering are:
  - *Aimed at large software*
  - *Systematic and well-defined techniques, methodologies and tools*
  - *Tools to design, code, test and maintain quality software*
  - *Applicable within a resource constrained environment*

Introduction

- Project:
  - Dictionary Definition – ***a planned activity***
    - *Being planned assumes that to a large extent we can determine how we are going to carry out a task before we start.*
  - A project starts from scratch with a definite mission, generates activities involving a variety of human and non-human resources .
  - All project activities are directed towards fulfilment of the mission and stop once the mission is fulfilled.

Introduction

- Project is Temporary:
  - This means that projects are not a permanent feature.
  - It doesn’t matter how long the project is, it has some starting date and some ending date.
  - It is important to mention here that its finish must b specified; otherwise it’s not a project

Introduction

- Project is Unique
  - Every project is unique.
  - No matter how similar a project is, still it is unique as there must be some difference
  - E.g. two similar towers (in shape) are to be constructed adjacent to each other.
  - Although it looks similar but still the area of construction is different, planned inhabitants of the buildings are different and so on.

Introduction

- Project must have a Deliverable :
  - All the projects must have a deliverable, means that it must have a target, objective that has to be achieved.
  - The deliverable can be a specified product (this can be a part of a bigger project), a service or a result.
  - The project is declared closed once the objectives of the project have been accomplished or it is felt that the deliverables of the project are no more needed.

Introduction

- ***Formally,***
  - ***A project is a***: Group of unique, inter-related activities that are planned and executed in a certain sequence to create a unique product or service, within a specific time frame, budget and the clients specifications.
- According to **Project Management Institute (PMI)**
  - Project is a temporary endeavor undertaken to create a unique product or service.
- The **British Standard** defines a project as:
  - A unique set of coordinated activities, with definite starting and finishing points, undertaken by an individual or organization to meet specific objectives within defined schedule, cost and performance parameters".

Some Project Initiatives

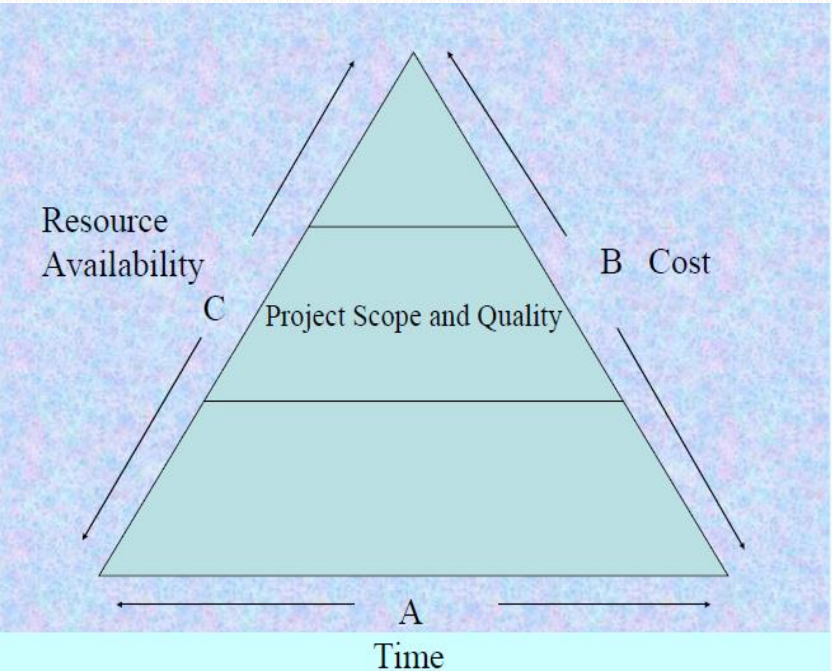
- Redesigning or relocating a production facility *(Manufacturing Project)*
- Implementing a management information system *(MIS Project)*
- Developing a new alloy required for a space vehicle. *(Spacecraft Project)*
- Constructing national highways *(Infrastructure Project)*
- Organizing the Olympics *(Sports project)*
- Constructing a dam for better irrigation facilities. *(Infrastructure Project)*
- Launching of new product *(Advertising and Marketing Project)*
- Implementing a new computer system *(IT project or upgrade)*
- Designing and Implementing a new organizational structure *(HR Project)*
- Designing and Constructing a house or colony *(Construction Project)*

Project Parameters

- Some common constraints that influence a project are:
  - *Scope*
  - *Quality*
  - *Cost*
  - *Time*
  - *Resources*

Project Parameters

- The scope and the quality of the project is influenced by variety of constraints like:
  - Time
  - Cost and
  - Availability of resources



Scope Triangle

Project Objectives

- Project objectives is defined as the general statement of desired output or end result of the project.
- The objectives of any project must be:
  - Specific
  - Measurable
  - Achievable
  - Relevant
  - Time Bounded
- These objective are aggregated as SMART

Project Objective

- Specific:
  - The word “specific” means to clearly defines the goals, in detail, without leaving room for wrong interpretations.
  - Consider the so-called Five W’s, where the five W stand for the following questions: Who? What? When? Where? Why?
  - Specific means, for instance, that you must be able to answer all these five questions.
- Measurable:
  - The word “measurable” refers to the measures and specifications of the performances that will determine if the goal has been achieved.
  - Measurable goals answer questions like “How much?” or “How many?”

Project Objective

- Achievable:
  - The word “achievable” refers to the fact that the team has a reasonable expectation of completion with success.
  - Every project manager should give achievable goals to his collaborators.
  - If they work on extremely difficult jobs, their productivity can be reduced and this could have a negative effect on motivation.
  - The feeling of never being able to achieve the project goal within the deadline can bring the project to failure.

Project Objective

- Relevant:
  - A goal is “relevant” when it is in line with group or business goals.
  - If the goal of the project has no relevance to the general vision, it is practically useless and cannot be defined as a SMART goal.
- Time Bounded:
  - The last feature of a SMART goal is “having a deadline”.
  - It is necessary to include a project deadline or the specific period within which the goals has to be achieved.
  - A deadline helps to create the needed urgency and helps to responsible stakeholders to focuses on the commitments they have made.

Classification of Projects

- Every Project is different from one another & can be classified based on several different points.
- The classification of projects in project management varies according to different factors such as objectives, source of capital, its content and so on.
- Projects can be classified based on the following factors:
    - According to Source of Capital
    - According to its Content.
    - According to its Objectives.
    - According to Involvement

According to Source of Capital

Project can be classified on the basis of Source of Capital as:

- Public:
  - Financing comes from Governmental institutions.
- Private:
  - Financing comes from businesses or private incentives.
- Mixed:
  - Financing comes from a mixed source of both public and private funding.

According to Project Content

Project can be classified on the basis of content as:

- Construction
  - These are projects that have anything to do with the construction of civil or architectural work.
  - Furthermore, construction is an engineering project and the process of planning its execution must be done to achieve the desired outcome.
- IT
  - Any project that has to do with software development, IT system, etc. are IT projects.
  - The types of project management information systems vary across the board, but in today's world are very common.

According to Project Content

- Business
  - These projects are involved with the development of a business idea, management of a work team, cost management, etc., and they usually follow a commercial strategy.
- Service or Project Production
  - These are projects that involve the development of an innovative product or service, design of a new product, etc. They are often used in the R & D department.

According to its Involvement

- Departmental:
  - When a certain department or area of an organization is involved.
- Internal:
  - When a whole company itself is involved in the project's development.
- Matriarchal:
  - When there is a combination of departments involved.
- External:
  - When a company outsources external project manager or teams to execute the project.

According to its Objectives

- Production:
  - Oriented at the production of a product or service taking into consideration a certain determined objective to be met by an organization.
- Social:
  - Oriented at the improvement of the quality of life of people. This can be in the form of rendering corporate social responsibility (CSR) to the people.
- Educational:
  - Oriented at the education of others. This is always done to make them better.
- Community:
  - Oriented at people too, however with their involvement.
- Research:
  - Oriented at innovation and the gaining of knowledge to enhance the operational efficiency of an organization.

Software Project

- A software project is the complete, procedure of software development from requirement gathering to testing and maintenance, carried out in a specified period of time to achieve intended software product.
- Progress of software development is not obviously visible
- Modifications of software products are more easy and flexible
- Software products are usually more complex than the hardware products in terms of development or construction cost.

Software Project vs Other Projects

- The product of software project have certain characteristics which make them different.
- **Invisibility:**
  - *When a physical artifact such a bridge or road is being constructed the progress being made can actually be seen.*
  - *With software, progress is not immediately visible.*
  - *In Software Development, there is a level of uncertainty.*
  - *It isn't easy to accurately define detailed requirements for software projects before starting the development.*

Software Project vs Other Projects

- **Complexity:**
  - *Considering the fact of investment per dollar, pound or euro, software products contain more complexity than other engineered artifacts.*
  - *For example, in a bridge, there is a clear structural relationship between parts, whereas software component relationships are much more complicated.*
  - *We can't measure the complexity of a software project until we work on it.*

Software Project vs Other Projects

- **Conformity:**
  - *Physical systems are governed by consistent physical law, while software developers have to conform to the requirements of human clients.*
  - *Software engineers (developers) have to conform to the client or organizational requirements and if they are inconsistent on what they need then developing software can be a difficult job.*

Software Project vs Other Projects

- **Flexibility:**
  - *Software systems are particularly subject to change.*
  - *A bridge has to be built in a specific order, whereas we can make software much more flexibly and restructure parts quite freely.*

Software Project vs Other Projects

- Furthermore Software Projects are intangible, and managing a software project is more managing interpersonal communication and less about administration.
- Also, it would be best if you had a highly skilled team of developers, QA, and another stakeholder to develop a software project, unlike making a bridge where skills can be less or more.

Categories of Software Project

- Some of the categories are
  - Information System versus Embedded Systems
  - Outsources Projects
  - Compulsory versus voluntary users
  - Objective-driven Development
- ***Information system*** interfaces with the organization whereas an ***embedded system*** interfaces with a machine.
- Typical example for an information system can be inventory system maintained by an organization.
- An embedded system or an industrial system can be a process control system such as maintaining air conditioning equipment in a company.

Categories of Software Project

- Projects are also defined by producing a product or meeting the objectives such projects are object oriented project.
- Such project produces the product must meet all kinds of client requirements.
- In general, software projects have an objective-driven stage that recommends the new system to meet identified requirements and a project-driven stage to actually develop the product.

Project Manager

- A project manager is the person responsible for leading a project from its inception to execution.
- This includes planning, execution and managing the people, resources and scope of the project.
- The project manager has full responsibility and authority to complete the assigned project.

Skills of Project Manager

- Leadership
- Negotiation
- Scheduling
- Cost Control
- Risk Management
- Critical Thinking
- Communication
- Task management
- Quality Management
- Sense of Humor

Roles & Responsibilities

- Activity & Resource Planning
- Organizing & Motivating the Project team
- Controlling Time Management
- Cost Estimation & Developing the budget
- Ensuring the Customer satisfaction
- Analyzing and Managing Project Risk
- Managing reports & necessary Documentation

Project Management

- Managing a project is called Project Management.

According to PMI

“Project Management is the application of knowledge, skills, tools and techniques to project activities in order to meet or exceed stakeholder needs and expectations”.

Project Management includes:

- defining project goals,
- specifying how the goals will be accomplished,
- what resources are needed, and relating budgets and time for completion.

Project Management Objectives

- Coordinate the various interrelated processes of the project.
- Ensure project includes all the work required, and only the work required, to complete the project successfully.
- Ensure that the project is completed on time and within budget.
- Ensure that the project will satisfy the needs for which it was undertaken.
- Ensure the most effective use of the people involved with the project.
- Promote effective communication between the projects team members and key stakeholders.
- Ensure that project risks are identified, analyzed, and responded.

Why Project Management

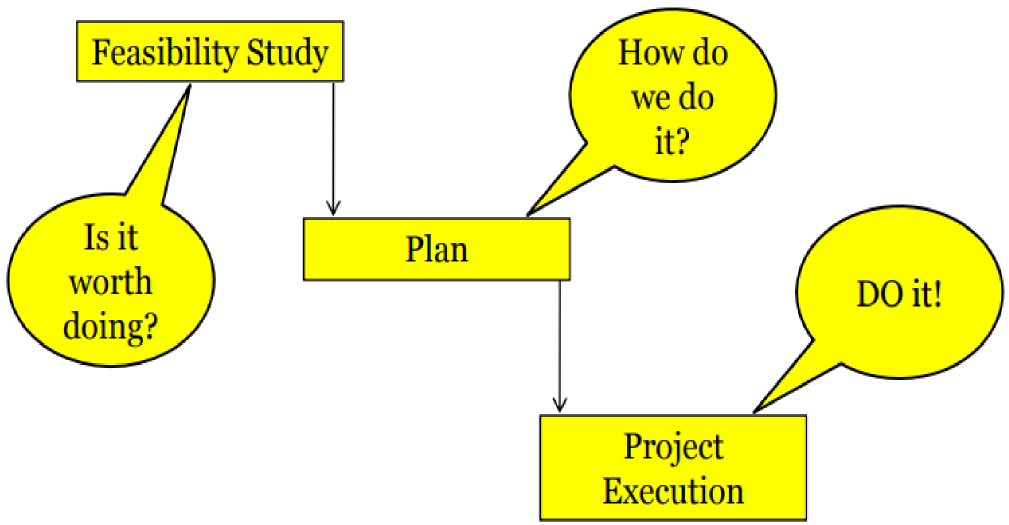
- The decision on whether or not to set up a separate project management division is subjective as it depends upon various factors some of them are:
  - *Interactions or interdependencies between various departments.*
  - *Sharing of common resources*
  - *The importance of the project to the organization*
  - *Size of the project*
  - *Degree of unfamiliarity with the work involved and its complexity*
  - *Changes in the market and*
  - *The reputation of the organization*

Advantages of Project Management

- A better efficiency in delivering services, which means you are going to work smarter and not harder.
- Cost flexibility, you only pays the hours spent on delivering services, less hours means lower cost.
- Objectivity, far from office politics, a project manager can give an idea of what is better for the company.
- Once the project manager has in this hands the work, you can focus on the searching of others providers

Activities Covered by Software Project Management (SPM)

- A software project is not only concerned with the actual writing of software.
- Usually there are three successive processes that bring a new system into being.
  - *Feasibility Study*
  - *Planning*
  - *Execution*



Activities Covered by SPM

- ***Feasibility Study***
  - *This investigates whether a prospective project is worth starting that it has a valid business case.*
  - *Information is gathered about the requirements of the proposed application.*
  - *The probable developmental and operational costs, along with the value of the benefits of the new system, will also have to be estimated.*

Activities Covered by SPM

- **A strategic planning for the following feasibility:**
  - *Technical : Hardware, Software Infrastructure ???*
  - *Economical – Investment vs Rate of Return???*
  - *Legal: Legal procedures, IT policy, Cyber law, Ethics ????*
  - *Operational – Will the system come into operation as per the requirement or not????*
  - *Time – Tentative time period of completion*

Activities Covered by SPM

- ***Planning***
  - *If the feasibility study produces results which indicate that the prospective project appears viable, then planning of the project can take place.*
  - *Formulate an outline plan for the whole project and a detailed one for the first stage.*
  - *For a large project, detailed planning is not done right at the beginning.*
  - *More detailed planning of the later stages would be done as they approached.*

Activities Covered by SPM

- **Contents list for a project plan**
  - *Introduction and background of project*
  - *Project Objectives*
  - *Constraints : these could be included with project objectives*
  - *Project products : both deliverable products that the client will receive and intermediate products.*
  - *Activities to be carried out*
  - *Resources to be used*
  - *Risks to the project*

Activities Covered by SPM

- ***Execution***
  - *After completion of the planning the project can now be executed.*
  - *The execution of a project often contains design and implementation sub-phases.*
  - ***Design** is thinking and making decisions about the precise form of the products that the project is to create.*
  - *In the case of software, this could relate to the external appearance of the software, that is, the user interface, or the internal architecture.*
  - ***Implementation** refers to the real time execution of the final product of the software project.*

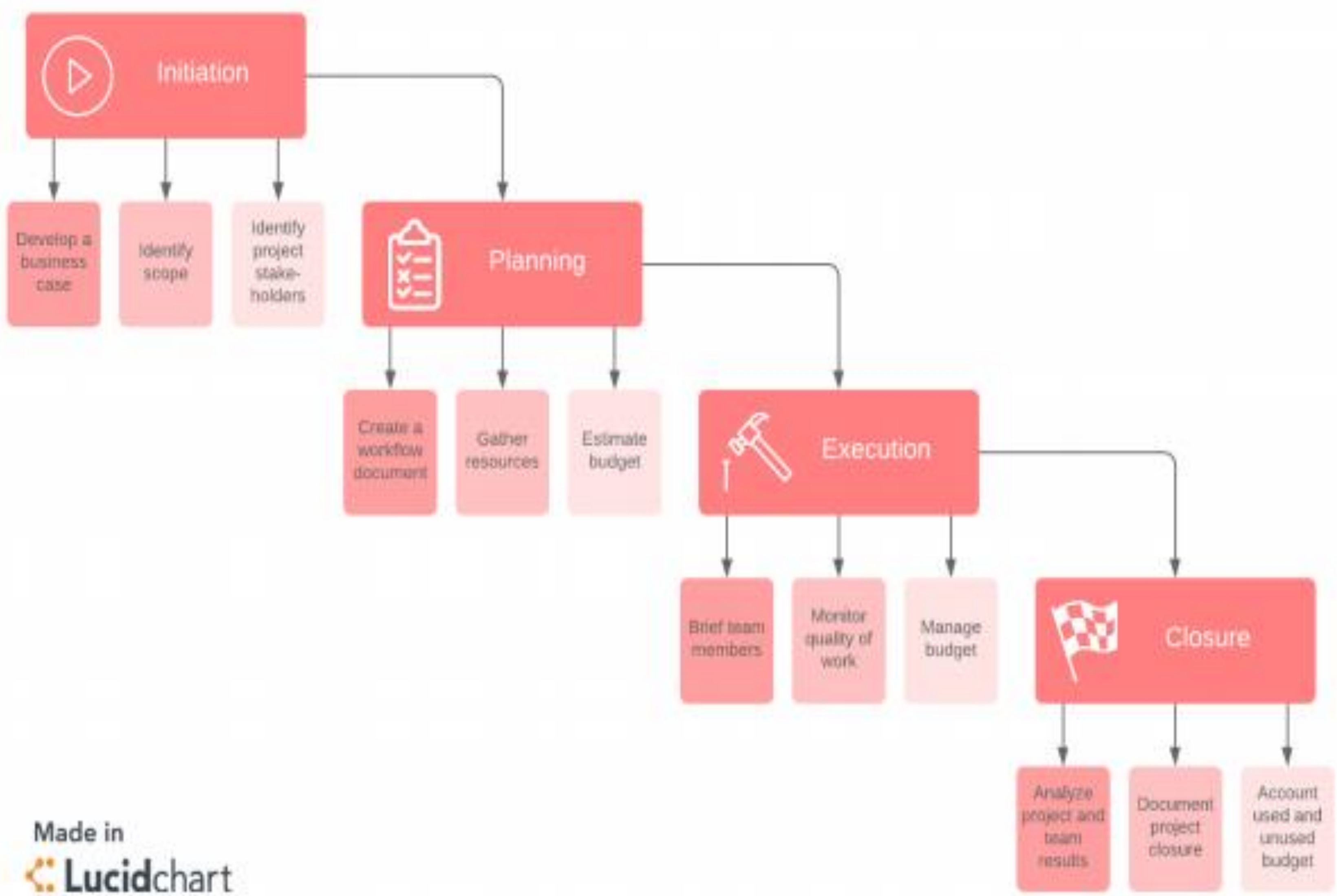
Project Management Life Cycle

- The project management life cycle describes high-level processes for delivering a successful project.
- It is one of the critical processes of any project.
- This is due to the fact that project lie cycle is the core process that connects all other project activities and processes together.
- When it comes to the activities of project management, there are many but basic activities done during the project will be discussed here.

Project Management Life Cycle

- The project management life cycle is usually broken down into four phases:
  - Initiation
  - Planning
  - Execution
  - Closure.
- These phases make up the path that takes your project from the beginning to the end.

  Note: Some methodologies also include a fifth phase—controlling or monitoring—but for our purposes, this phase is covered under the execution and closure phases.



Initiation
Develop a business case
Identify scope
Identify project stake-holders
Planning
Create a workflow document
Gather resources
Estimate budget
Execution
Brief team members
Monitor quality of work
Manage budget
Closure
Analyze project and team results
Document project closure
Account used and unused budget
Made in Lucidchart

Initiation

- Project initiation is the starting point of any project.
- In this process, first, you need to identify a business need, problem, or opportunity and all the activities related to winning a project takes place.
- During this period, the service provider proves the eligibility and ability of completing the project to the client and eventually wins the business.
- Secondly, brainstorm ways that your team can meet this need to solve this problem.

Initiation

- Then, it is the detailed requirements gathering which comes next.
- During the requirements gathering activity, all the client requirements are gathered and analysed for implementation.
- In this activity, negotiations may take place to change certain requirements or remove certain requirements altogether.
- Usually, project initiation process ends with requirements gathering.

Steps in Initiation Phase

- The initiation phase may include the following steps:
  - Undertaking a feasibility study: Identify the primary problem your project will solve and whether your project will deliver a solution to that problem.
  - Identifying scope: Define the depth and breadth of the project
  - Identifying deliverables: Define the product or service to provide
  - Identifying project stakeholders: Figure out whom the project affects and what their needs may be.

Steps in Initiation Phase

- Developing a business case: Use the above criteria to compare the potential costs and benefits for the project to determine if it moves forward.
- Developing a statement of work: Document the project's objectives, scope, and deliverables that you have identified previously as a working agreement between the project owner and those working on the project

Planning

- Project planning is one of the main project management processes.
- If the project management team gets this step wrong, there could be heavy negative consequences during the next phases of the project.
- In this process, the project plan is derived in order to address the project requirements such as, requirements scope, budget and timelines.
- Once the project plan is derived, then the project schedule is developed.

Planning

- Depending on the budget and the schedule, the resources are then allocated to the project.
- This phase is the most important phase when it comes to project cost and effort.
- Furthermore during this phase of the project management life cycle, we break down the larger project into smaller tasks, build our team, and prepare a schedule for the completion of assignments.

Steps for the Planning Phase

- Creating a project plan
  - Identify the project timeline, including the phases of the project, the tasks to be performed, and possible constraints
- Creating workflow diagrams
  - Visualize your processes using swim lanes to make sure team members clearly understand their role in a project
- Estimating budget and creating a financial plan
  - Use cost estimates to determine how much to spend on the project to get the maximum return on investment

Steps for the Planning Phase

- Gathering resources
  - Build your functional team from internal and external talent pools while making sure everyone has the necessary tools (software, hardware, etc.) to complete their tasks
- Anticipating risks and potential quality roadblocks
  - Identify issues that may cause your project to stall while planning to mitigate those risks and maintain the project's quality and timeline .

Execution

- The execution phase turns your plan into action.
- The project manager's job in this phase of the project management life cycle is to keep work on track, organize team members, manage timelines, and make sure the work is done according to the original plan.
- During the project execution, there are many reporting activities to be done.
- The senior management of the company will require daily or weekly status updates on the project progress.

Execution

- Furthermore, the client may also want to track the progress of the project.
- During the project execution, it is a must to track the effort and cost of the project in order to determine whether the project is progressing in the right direction or not.

Steps in Execution Phase

- Creating tasks and organizing workflows
  - Assign granular aspects of the projects to the appropriate team members, making sure team members are not overworked
- Briefing team members on tasks
  - Explain tasks to team members, providing necessary guidance on how they should be completed, and organizing process-related training if necessary
- Communicating with team members, clients, and upper management
  - Provide updates to project stakeholders at all levels.

Steps in Execution Phase

- Monitoring quality of work
  - Ensure that team members are meeting their time and quality goals for tasks
- Managing budget
  - Monitor spending and keeping the project on track in terms of assets and resources

Closure Phase

- Once your team has completed work on a project, we enter the closure phase.
- In the closure phase, we provide final deliverables, release project resources, and determine the success of the project.
- Just because the major project work is over, that doesn’t mean the project manager’s job is done, there are still important things to do, including evaluating what did and did not work with the project.

Steps in Closure Phase

- Analyzing project performance
  - Determine whether the project's goals were met (tasks completed, on time and on budget) and the initial problem solved using a prepared checklist.
- Analyzing team performance
  - Evaluate how team members performed, including whether they met their goals along with timeliness and quality of work.
- Documenting project closure
  - Make sure that all aspects of the project are completed with no loose ends remaining and providing reports to key stakeholders .

Steps in Closure Phase

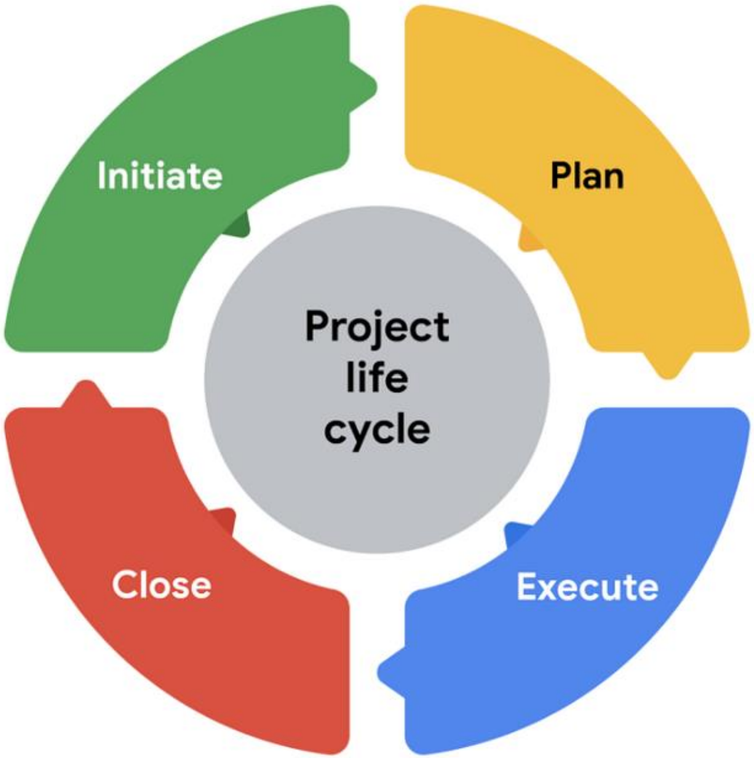
- Conducting post-implementation reviews
  - Conduct a final analysis of the project, taking into account lessons learned for similar projects in the future.
- Accounting for used and unused budget
  - Allocate remaining resources for future projects

SPM Framework

- Software Project Management Framework consists of three parts:
  - Project Lifecycle
  - Project Control Cycle
  - Tools and Templates

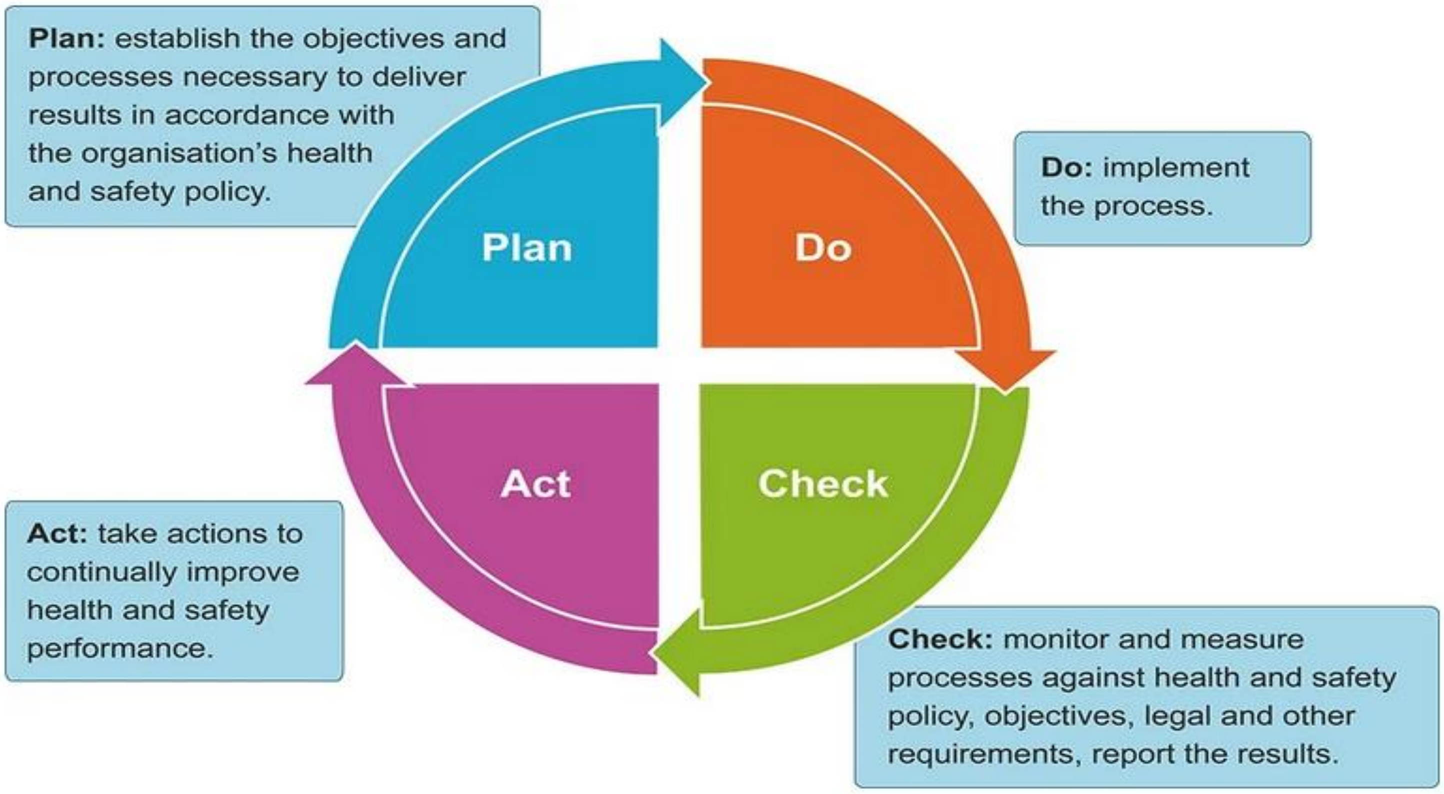
SPM Framework

Project Lifecycle ***(Similar to Project Management Life Cycle)***



SPM Framework

- ***Project Control Cycle***



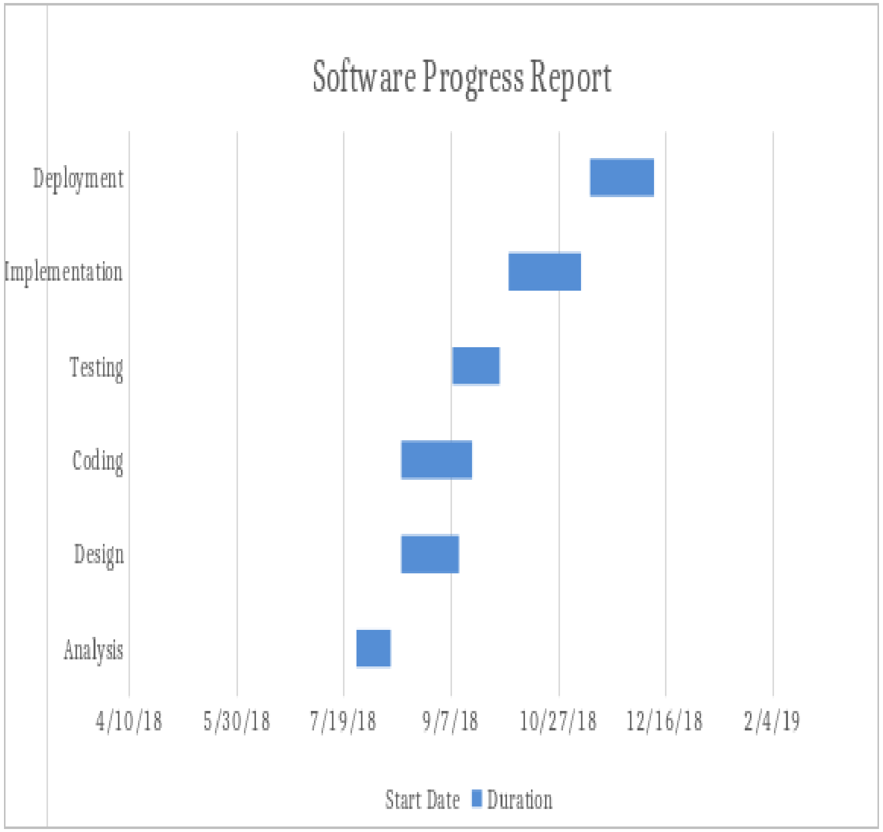
SPM Framework

Tools and Templates

- To manage the project management system adequately and efficiently, we use project management tools
- Some tools are:
  - *Gantt Chart*
  - *PERT Chart*
  - *Work Breakdown Structure*

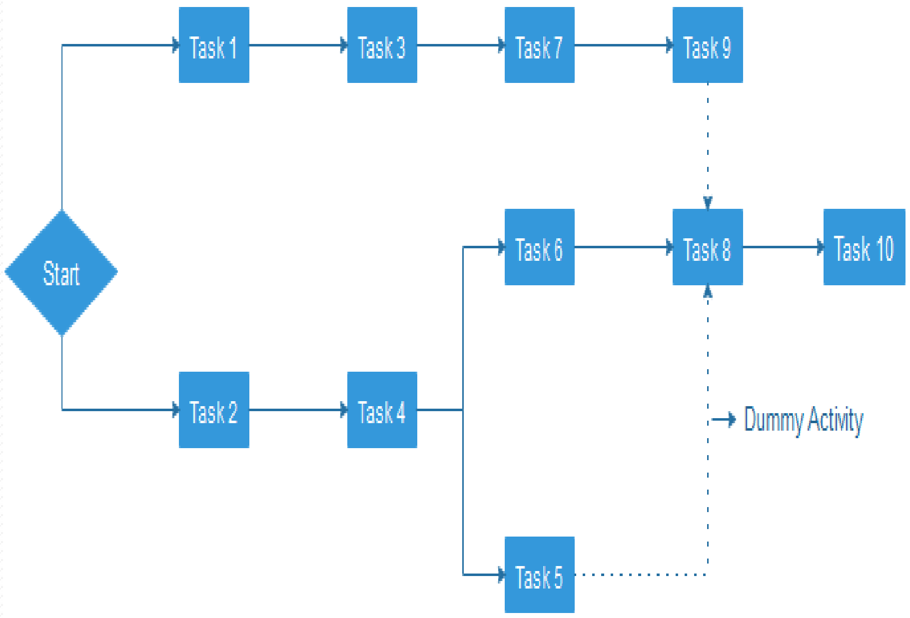
SPM Framework

- ***Gantt Chart*** *(Schedule Of Project)*
  - It is one of the most popular and helpful ways of showing activities displayed against time.
  - Each activity represented by a bar.



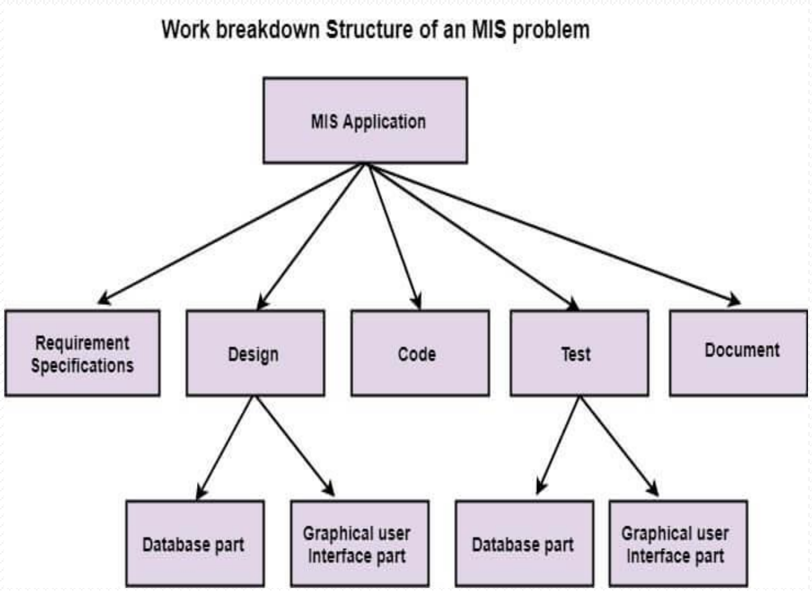
SPM Framework

- ***PERT Chart***
  - PERT stands for Program Evaluation Review Technique.
  - In Project Management, PERT chart represented as a network diagram concerning the number of nodes, which represents events.
  - The direction of the lines indicates the sequence of the task.



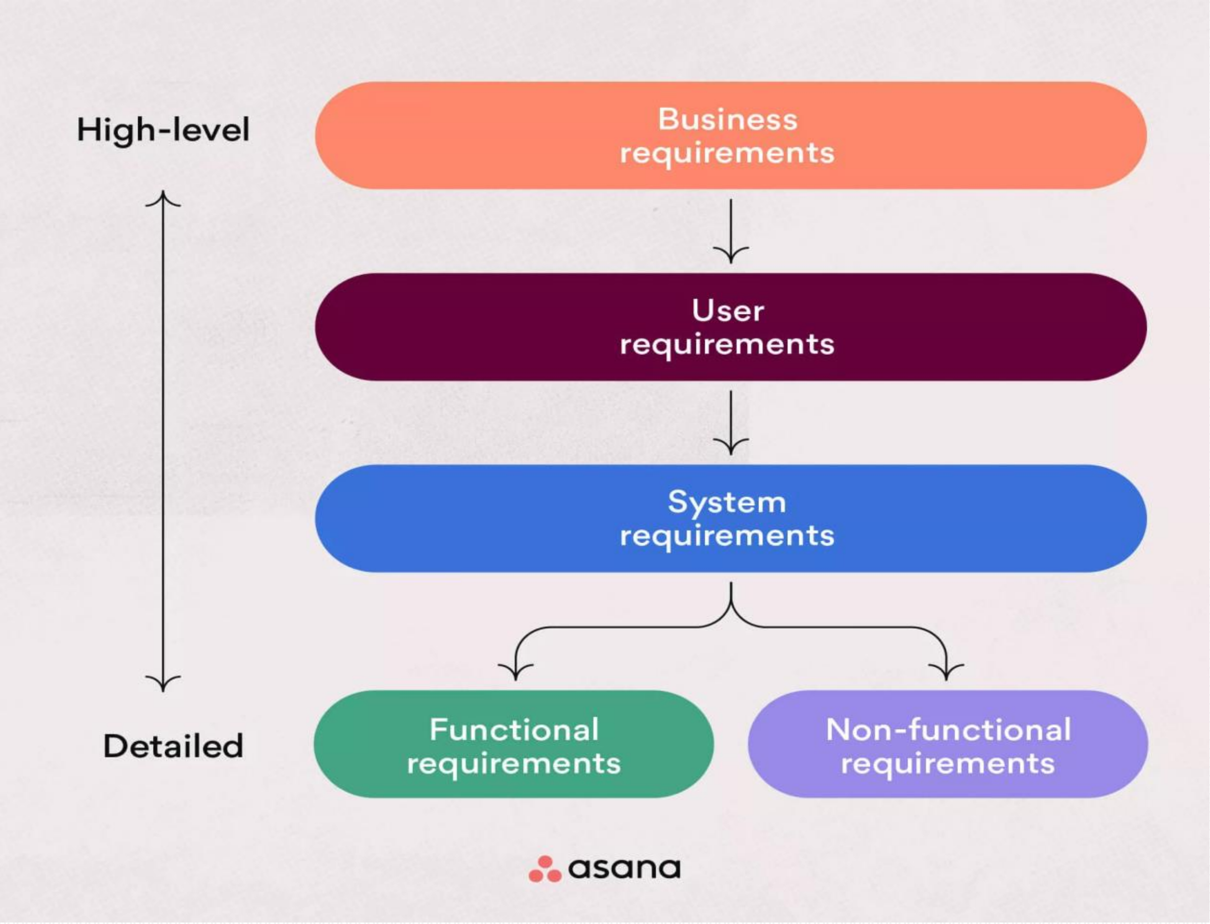
SPM Framework

- ***Work Breakdown Structure***
  - Product Breakdown Structure (BBS) is a management tool and necessary a part of the project designing.
  - It's a task-oriented system for subdividing a project into product parts.
  - The product breakdown structure describes subtasks or work packages and represents the connection between work packages.



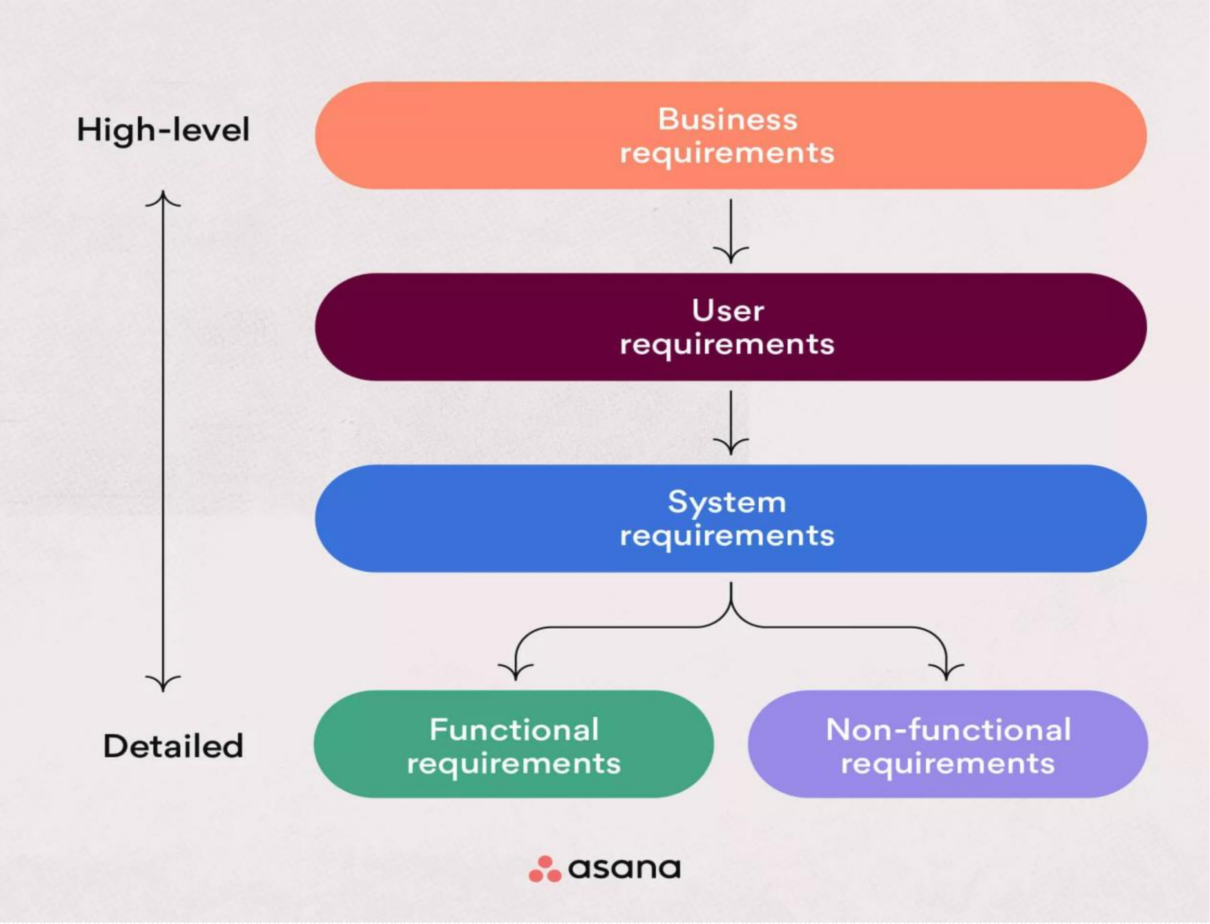
SPM Framework

- ***Templates*** ***(For Business Requirements)***



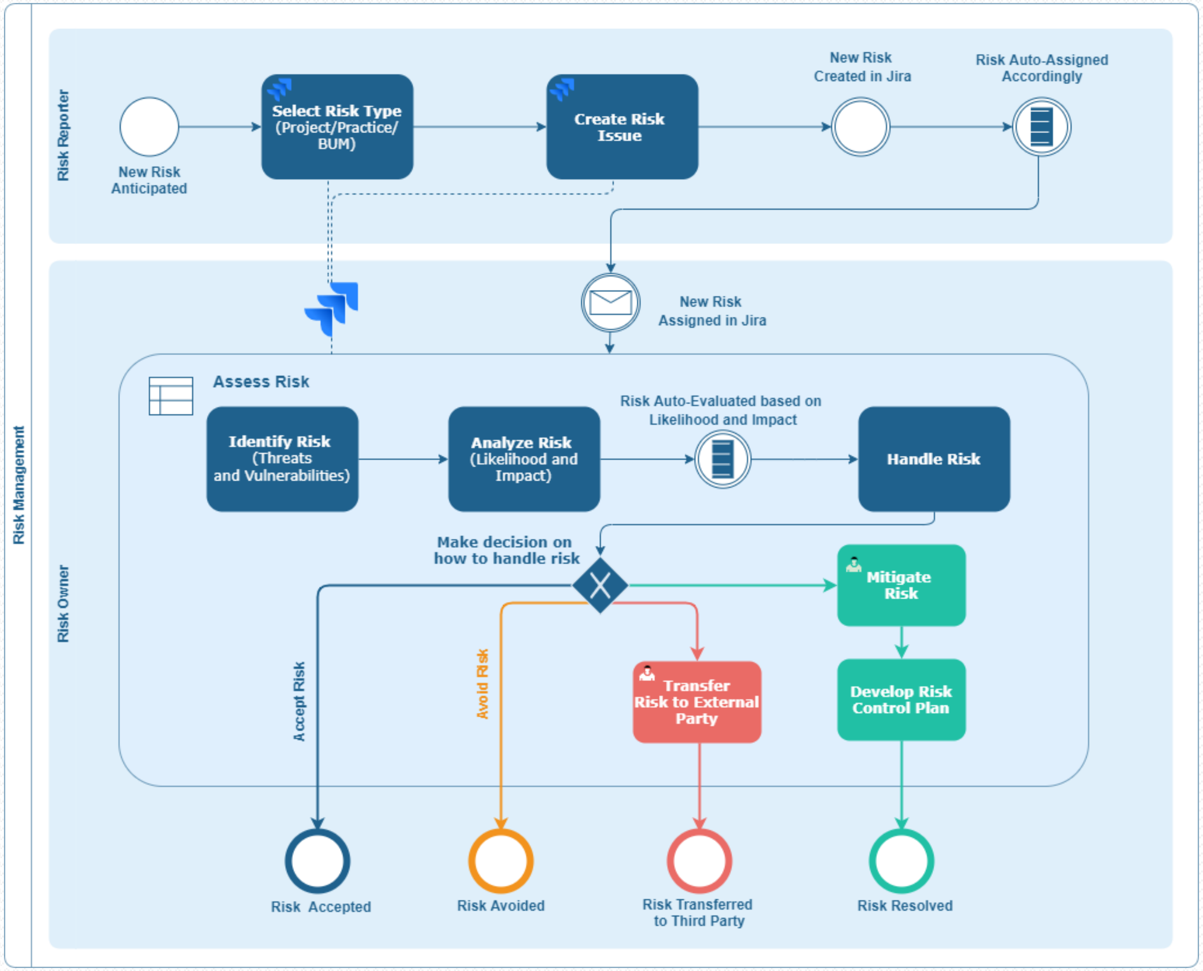
SPM Framework

- ***Templates*** ***(For Business Requirements)***



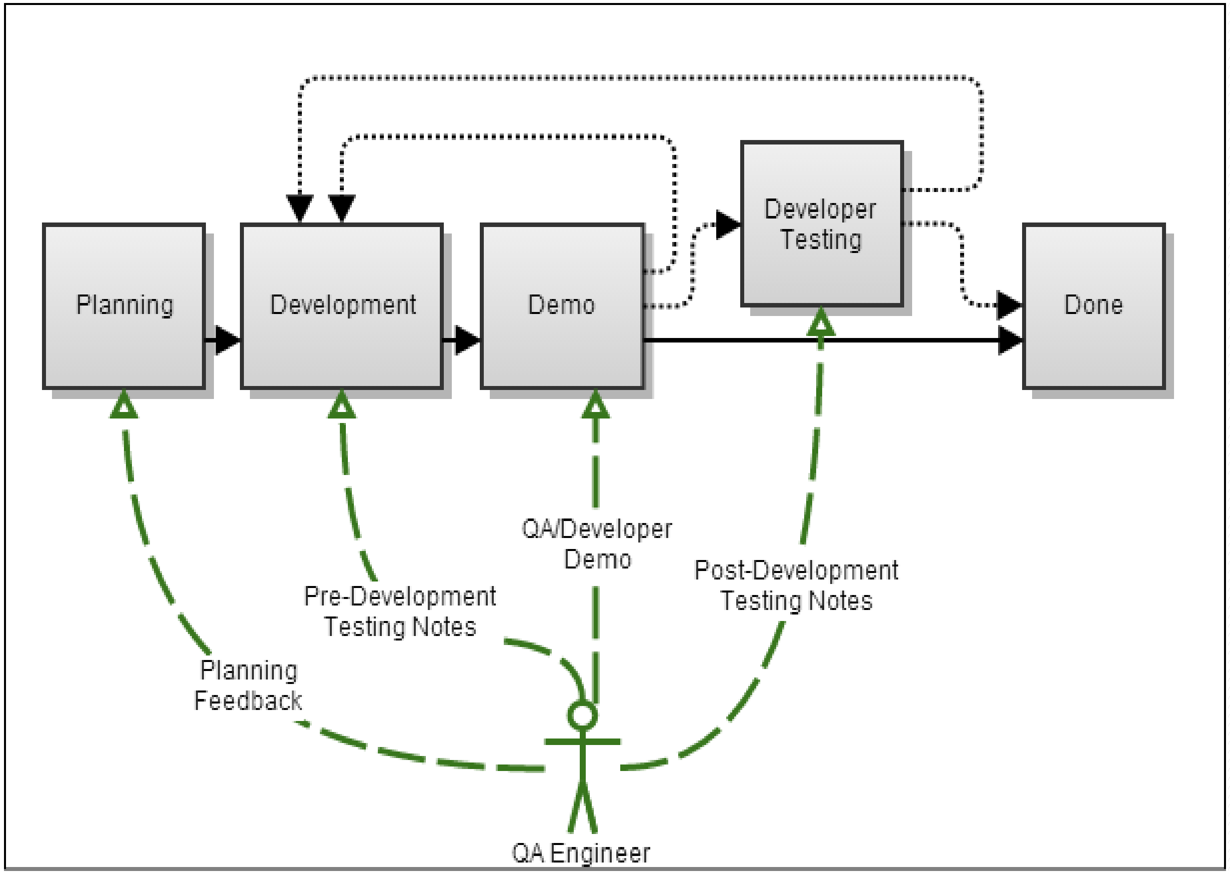
SPM Framework

- ***Templates*** ***(For Risk Management)***



SPM Framework

- ***Templates*** ***(For Quality Assurance)***



Project Plan

- The project planning process is the main tool used to ensure that tasks are completed in timely manner.
- A project may best be defined as a venture taken to ensure that a deliverable is completed within a specific timeframe and that certain criteria or objectives are met.
- In order to make certain that a project has the best chance of success and risk of failure has been minimized, a plan is devised to determine the most effective strategy for completion.

Types of Project Plan

Plan	Description
Quality plan	Describes the quality procedures and standards that will be used in a project.
Validation plan	Describes the approach, resources and schedule used for system validation.
Configuration management plan	Describes the configuration management procedures and structures to be used.
Maintenance plan	Predicts the maintenance requirements of the system, maintenance costs and effort required.
Staff development plan	Describes how the skills and experience of the project team members will be developed.

Project Plan Control

PLAN MONITOR CONTROL CYCLE

Enter your sub headline here

	Plan	Monitor	Control
Define	The project goals and objectives are defined, and the strategies to achieve them are developed. The information flow in this stage is as follows.	The progress of the project is monitored, and the performance is measured against the planned objectives. The information flow in this stage is as follows.	Corrective actions are taken to bring the project back on track if there are any deviations from the plan. The information flow in this stage is as follows.
Input	Project requirements, stakeholder expectations, and project scope statement	Project plan, project schedule, and project budget	Project status report, change requests, and corrective actions
Output	Project plan, project schedule, and project budget	Project status report, change requests, and corrective actions	Approved change requests, updated project plan, and updated project schedule.
Tools and Techniques	Project management methodologies, expert judgement, and project management software	Performance measurement techniques, progress reporting, and project management software	Change control processes, configuration management, and project management software

End of Chapter 1

Thank You !!!!